

AP Statistics FRQ Practice Exam 06

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AP Statistics FRQ Practice Exam 6

APSTAT-FRQ6-001

Section: FRQ

Type: frq

Primary Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Secondary Unit: Unit 4, Inference for Quantitative Data: Means

FRQ Type: Formulate Questions and Collect Data

Prompt:

A school district wants to estimate the mean one-way commute time for all 12,400 tenth-grade students enrolled in its 40 high schools. The district randomly selects 8 of the 40 schools. At each selected school, it randomly selects 2 tenth-grade homerooms and then randomly selects 5 students from each chosen homeroom. The 80 sampled students report their commute times.

Selected school	A	B	C	D	E	F	G	H
Mean commute time in sample (minutes)	18	22	25	27	31	34	29	23

- (a) Identify the population, sample, and population parameter of interest.
- (b) Describe the sampling method, including each stage of random selection.
- (c) Because 10 students were sampled from each selected school, calculate the overall sample mean commute time. Interpret this statistic in context.
- (d) State a statistical question the district can investigate with the study. Identify the observational unit and variable measured.
- (e) School enrollment, number of homerooms, and homeroom size vary substantially. Explain why the procedure may give different tenth-grade students different probabilities of selection. Describe one redesign or analysis adjustment that would better support an estimate for all 12,400 students.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Identifies the population as all 12,400 tenth-grade students in the district and the sample as the 80 selected students from the 8 selected schools.
- Point 2: Identifies μ as the true mean one-way commute time, in minutes, for all 12,400 tenth-grade students.

(b) 2 points

- Point 1: Identifies the design as a multistage sample and correctly describes the first two stages: randomly selecting schools and then homerooms within those schools.
- Point 2: Describes the final random selection of 5 students within each chosen homeroom, rather than treating every student in a chosen homeroom as sampled.

(c) 2 points

- Point 1: Calculates $\bar{x} = (18 + 22 + 25 + 27 + 31 + 34 + 29 + 23)/8 = 26.125$ minutes, using equal weights because each selected school contributes 10 students.
- Point 2: Interprets 26.125 minutes as the average reported one-way commute time among the 80 sampled students, not automatically as the exact population mean.

(d) 2 points

- Point 1: States an appropriate statistical question, such as, “What is the mean one-way commute time for tenth-grade students in the district?”
- Point 2: Identifies an individual tenth-grade student as the observational unit and that student's reported one-way commute time in minutes as the quantitative variable.

(e) 2 points

- Point 1: Explains that selecting the same number of schools, homerooms, and students at each stage does not give every student the same inclusion probability when schools have different numbers of homerooms or homerooms have different enrollments.
- Point 2: Proposes a valid improvement, such as taking an SRS from a complete districtwide student roster, selecting schools or homerooms with probabilities proportional to enrollment, or applying weights based on known inclusion probabilities.

Model Guidance:

The target is the districtwide mean for all tenth-grade students, while the sample contains 80 students reached through three random stages. The design first samples schools, then homerooms within schools, then students within homerooms. Since each school contributes 10 students, the unweighted sample mean is the average of the eight school-specific sample means: 26.125 minutes.

Random selection at every stage makes this a probability sample, but it does not guarantee equal student probabilities. A student in a school with few homerooms or in a small selected homeroom may have a greater chance of selection than a student in a large school or class. An SRS from a complete roster is the simplest equal-probability redesign. Proportional selection or appropriate sampling weights can also align the estimate with the district's student population.

Expected Points:

10

APSTAT-FRQ6-002

Section: FRQ

Type: frq

Primary Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Secondary Unit: None

FRQ Type: Analyze Data, One- or Two-Variable Data and Probability

Prompt:

An archive box contains 12 film reels, of which 3 have a known chemical deterioration problem and 9 do not. An archivist randomly selects 4 different reels without replacement for immediate inspection.

- (a) Explain why the number of deteriorated reels selected does not have a binomial distribution.
- (b) Calculate the probability that none of the 4 selected reels is deteriorated. Then calculate the probability that at least 1 selected reel is deteriorated.
- (c) Calculate the probability that exactly 2 of the 4 selected reels are deteriorated. Interpret the probability in context.
- (d) Given that the first selected reel is not deteriorated, calculate the probability that the second selected reel is deteriorated. Use the result to explain why the selection outcomes are dependent.
- (e) Describe a random-digit simulation that could estimate the probability that at least 1 of 4 selected reels is deteriorated. Include how one trial is conducted and how the probability is estimated.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Notes that 4 reels are selected without replacement from a small collection, so the outcome of one draw changes the composition for later draws.
- Point 2: States that independence and a constant success probability are violated, so a binomial model is not appropriate.

(b) 2 points

- Point 1: Calculates $P(\text{no deteriorated reels}) = C(9, 4)/C(12, 4) = 126/495 \approx 0.2545$.
- Point 2: Calculates $P(\text{at least 1 deteriorated reel}) = 1 - 126/495 = 369/495 \approx 0.7455$.

(c) 2 points

- Point 1: Calculates $P(\text{exactly 2 deteriorated reels}) = [C(3, 2)C(9, 2)]/C(12, 4) = 108/495 \approx 0.2182$.
- Point 2: Interprets the result: in repeated random selections of 4 reels from this box, about 21.8% of the selections would contain exactly 2 deteriorated reels.

(d) 2 points

- Point 1: Calculates $P(\text{second deteriorated} \mid \text{first not deteriorated}) = 3/11 \approx 0.2727$.
- Point 2: Compares $3/11$ with the initial probability $3/12 = 0.25$ and explains that the first result changes the probability on the second draw, demonstrating dependence.

(e) 2 points

- Point 1: Gives a valid simulation method, such as labeling the reels 01 through 12 with 01–03 deteriorated, reading random two-digit numbers, ignoring 00, 13–99, and repeats, and continuing until 4 distinct labels form one trial.
- Point 2: Repeats many trials, records whether each trial contains at least one of 01–03, and estimates the probability with the proportion of trials having that event.

Model Guidance:

Without replacement, the chance of drawing a deteriorated reel changes after every draw. Exact probabilities can be obtained by counting equally likely four-reel subsets. There are $C(12, 4) = 495$ possible subsets. Of these, $C(9, 4) = 126$ contain no deteriorated reel, so the complement probability for at least one is about 0.7455. Exactly two requires choosing two of the three deteriorated reels and two of the nine others, giving $108/495 \approx 0.2182$.

After a nondeteriorated first reel, all three deteriorated reels remain among only 11 total reels, so the next-draw probability rises from $3/12$ to $3/11$. A simulation must preserve this no-replacement structure by rejecting duplicate labels within a trial. The long-run fraction of simulated four-reel samples containing a deteriorated label estimates the desired probability.

Expected Points:

10

APSTAT-FRQ6-003

Section: FRQ

Type: frq

Primary Unit: Unit 3, Inference for Categorical Data: Proportions

Secondary Unit: Unit 1, Exploring One-Variable Data and Collecting Data

FRQ Type: Analyze Data, Statistical Inference (Confidence Interval or Significance Test)

Prompt:

A restaurant-industry group studies the use of reusable takeout containers in two regions. Independent simple random samples are taken from restaurants in each region. Each region contains more than 2,000 restaurants.

Region	Sample size	Offers reusable containers
Coastal	180	117
Inland	160	88

Construct and interpret a 95% confidence interval for $p_C - p_I$, where p_C and p_I are the true proportions of all restaurants in the coastal and inland regions, respectively, that offer reusable takeout containers. Use $z^* = 1.96$.

(a) Define the population parameters and identify the appropriate confidence-interval procedure.

(b) Check the conditions for the confidence-interval procedure.

(c) Calculate the sample difference, standard error, and margin of error, and construct the confidence interval.

(d) Interpret the confidence interval in context.

(e) An industry representative claims that the coastal-region proportion is higher than the inland-region proportion. Use the interval to evaluate the claim.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Defines p_C and p_I as the true proportions of all restaurants in the coastal and inland regions, respectively, that offer reusable takeout containers.
- Point 2: Selects an independent two-sample z confidence interval for a difference in population proportions.

(b) 3 points

- Point 1: Cites the independent SRSs and the independence of the two regional samples.
- Point 2: Verifies the 10% condition because 180 and 160 are each less than 10% of the corresponding region's population of more than 2,000 restaurants.
- Point 3: Verifies the large-counts condition: coastal successes and failures are 117 and 63, and inland successes and failures are 88 and 72, all at least 10.

(c) 3 points

- Point 1: Calculates $\hat{p}_C = 117/180 = 0.650$, $\hat{p}_I = 88/160 = 0.550$, and $\hat{p}_C - \hat{p}_I = 0.100$.
- Point 2: Calculates $SE = \sqrt{[(0.65)(0.35)/180 + (0.55)(0.45)/160]} \approx 0.05302$ and margin of error $1.96(0.05302) \approx 0.1039$.
- Point 3: Constructs 0.100 ± 0.1039 , giving the interval $(-0.0039, 0.2039)$, allowing for minor rounding differences.

(d) 1 point

- Point 1: States that the group is 95% confident that the coastal proportion minus the inland proportion is between about -0.4 and 20.4 percentage points.

(e) 1 point

- Point 1: States that the interval does not provide convincing evidence that the coastal proportion is higher because the interval includes 0 and a small negative difference.

Model Guidance:

This is a two-proportion estimation problem with independent random samples. For a confidence interval, use separate sample proportions in the standard error rather than a pooled proportion. All success and failure counts exceed 10, and the sample sizes are below 10% of the regional populations.

The observed difference is $0.650 - 0.550 = 0.100$. The unpooled standard error is about 0.05302, giving a 95% margin of error of about 0.1039 and the interval $(-0.0039, 0.2039)$. The data are compatible with anything from a very small inland advantage to a coastal advantage of about 20.4 percentage points. Because 0 is plausible, the sample does not establish the representative's directional claim at the corresponding two-sided 5% level.

Expected Points:

10

APSTAT-FRQ6-004

Section: FRQ

Type: frq

Primary Unit: Unit 5, Regression Analysis

Secondary Unit: Unit 1, Exploring One-Variable Data and Collecting Data

FRQ Type: Interpret Results, Integrated Analysis and Justification

Prompt:

An audio-restoration studio records x , the duration of an archival recording in minutes, and y , the technician's editing time in hours, for the first 8 projects completed during one month.

Project	1	2	3	4	5	6	7	8
Recording duration, x	20	30	40	50	60	70	80	90
Editing time, y	1.4	3.4	1.9	4.7	3.1	5.9	4.3	7.0

Computer output for predicting editing time from recording duration is

$$\hat{y} = 0.420 + 0.0644x$$

$$r = 0.826$$

$$r^2 = 0.683$$

Computer output for predicting recording duration from editing time is

$$\hat{x} = 12.997 + 10.600y$$

- (a) On paper, sketch a scatterplot of editing time against recording duration. Label both axes and describe the form, direction, strength, and any unusual features of the association.
- (b) Interpret the slope of the first regression line. Interpret the intercept and explain whether it is meaningful in this context.
- (c) Let u be recording duration measured in hours, so $u = x/60$. Rewrite the first regression equation to predict editing time in hours from u . State what happens to r when x is converted from minutes to hours, and explain why.
- (d) A manager claims that doubling a recording's duration from 30 to 60 minutes doubles its predicted editing time. Use the first regression line to evaluate the claim.
- (e) The eight projects were the first projects completed during one month, rather than a random sample from all studio projects. Explain how this limits the scope of conclusions.
- (f) For a project with 4.5 hours of editing time, compare the recording-duration prediction obtained by algebraically solving the first line for x with the prediction from the second regression line. Explain which method is appropriate.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Plots the 8 ordered pairs with recording duration on the horizontal axis and editing time on the vertical axis, using suitable labels and scales.

- Point 2: Describes a moderately strong, positive, roughly linear association with noticeable scatter and no single clearly isolated point.

(b) 2 points

- Point 1: Interprets the slope: among these projects, each additional minute of recording duration is associated with an increase of about 0.0644 hour in predicted editing time, on average.

- Point 2: Interprets 0.420 hour as the predicted editing time for a zero-minute recording and explains that it is not meaningful because zero is outside the observed 20-to-90-minute range and is not a substantive project.

(c) 2 points

- Point 1: Substitutes $x = 60u$ to obtain $\hat{y} = 0.420 + 3.864u$.

- Point 2: States that r remains 0.826 because multiplying a variable by a positive constant changes its units but not the standardized strength or direction of the linear association.

(d) 1 point

- Point 1: Calculates predictions of $0.420 + 0.0644(30) = 2.352$ hours and $0.420 + 0.0644(60) = 4.284$ hours, then rejects the claim because 4.284 is not twice 2.352; the nonzero intercept makes the fitted relationship nonproportional.

(e) 1 point

- Point 1: Explains that a convenience sample from one month may not represent other projects, technicians, workloads, or time periods, so the fitted relationship should not be generalized broadly without additional random or representative sampling.

(f) 2 points

- Point 1: Calculates the algebraic inverse prediction $x = (4.5 - 0.420)/0.0644 \approx 63.4$ minutes and the correct reverse-regression prediction $\hat{x} = 12.997 + 10.600(4.5) \approx 60.7$ minutes.

- Point 2: Selects the second regression line because least-squares regression depends on which variable is the response; algebraically reversing the y-on-x line does not produce the x-on-y least-squares line unless the correlation is perfect.

Model Guidance:

The scatterplot rises overall but shows meaningful vertical scatter, consistent with $r = 0.826$. The first slope corresponds to about 0.0644 editing hour per additional recording minute. Its intercept describes a nonexistent zero-minute project and should not be given practical meaning.

If duration is expressed in hours, one unit of u equals 60 minutes, so the slope becomes $60(0.0644) = 3.864$ while the intercept stays 0.420. Correlation is unchanged by a positive unit conversion. The fitted values at 30 and 60 minutes are 2.352 and 4.284 hours, so doubling x does not double $\hat{y}$ when the intercept is nonzero.

For reverse prediction, algebraic inversion gives about 63.4 minutes, whereas the least-squares line constructed with duration as the response gives about 60.7 minutes. Use the latter because it minimizes squared errors in duration. Any conclusion remains descriptive of the convenience sample unless broader sampling supports generalization.

Expected Points:

10

SERIES DO-NOT-REPEAT BANK UPDATE, APSTAT-FRQ6

- APSTAT-FRQ6-001: Unit 1 primary; Unit 4 secondary; three-stage district sample selecting 8 of 40 high schools, 2 tenth-grade homerooms per school, and 5 students per homeroom from 12,400 students; table of eight equally weighted school sample means 18, 22, 25, 27, 31, 34, 29, and 23 minutes; identify population/sample/ μ /unit/variable, formulate a mean-commute question, calculate $\bar{x} = 26.125$, and diagnose unequal student inclusion probabilities caused by varying school and class sizes, with roster-SRS, proportional-selection, or weighting remedies; targets assuming random selection at several stages automatically makes the unweighted estimator self-weighting.
- APSTAT-FRQ6-002: Unit 2 primary; no secondary unit; without-replacement selection of 4 from 12 archival film reels, 3 deteriorated and 9 not; explain why the count is not binomial, use combinations to calculate $P(0) = 126/495$, $P(\text{at least } 1) = 369/495$, and $P(\text{exactly } 2) = 108/495$, use 3/11 versus 3/12 to show dependence, and design a distinct-label random-digit simulation; targets applying binomial probabilities to dependent draws and allowing repeated labels in a no-replacement simulation.
- APSTAT-FRQ6-003: Unit 3 primary; Unit 1 secondary; independent two-proportion z confidence interval comparing reusable-takeout-container availability in coastal and inland restaurant populations, using SRS results 117/180 and 88/160; check independent SRS/10%/four large-count conditions, calculate unpooled $SE \approx 0.05302$ and 95% interval $(-0.0039, 0.2039)$ for $p_C - p_I$, interpret direction and scope, and reject a claimed coastal advantage because zero is included; targets pooling proportions for an interval and treating a positive point estimate as established population ordering.
- APSTAT-FRQ6-004: Unit 5 primary; Unit 1 secondary; convenience sample of eight audio-restoration projects with raw duration/editing-time table, forward regression $\hat{y} = 0.420 + 0.0644x$, $r = 0.826$, $r^2 = 0.683$, and reverse regression $\hat{x} = 12.997 + 10.600y$; draw and describe a scatterplot, interpret slope and unusable intercept, convert duration from minutes to hours while preserving r , disprove a doubling/proportionality claim, limit generalization from one month's first projects, and compare algebraic inversion with the correct reverse least-squares prediction at $y = 4.5$; targets assuming transformed units change correlation, treating a nonzero-intercept model as proportional, and reversing a regression line algebraically.